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SECTION 1: ACTUATOR TORQUE MARGINS & TIMING CRITERIA

- 1.1 Motor-Operated Valve (MOV) Torque Margin:
- Required Available Torque Margin: $\geq 30\%$ above unseating breakaway torque.
 - **Warning Threshold: Torque margin between 20% and 29%.**
 - **Critical Threshold: Margin $< 20\%$ requires immediate actuator overhaul.**
- 1.2 Full Stroke Time Tolerance:
- Emergency Isolation Valves: Must fully close within 25.0 seconds.
 - Stroke time exceeding 30 seconds triggers safety loop compliance violation.

SECTION 2: SEAT LEAKAGE & ACOUSTIC CAVITATION STANDARDS

- 2.1 Class VI Shut-Off Criteria: Zero visible bubble leakage at ambient test conditions.
- 2.2 Acoustic Cavitation Thresholds (Ultrasonic 30 - 45 kHz band):
- Normal Non-Cavitating Operation: < 55 dB.
 - Mild Incipient Cavitation: 55 to 68 dB.
 - Severe Cavitation / Trim Erosion Risk: > 72 dB.

SECTION 3: PACKING GLAND & HIGH-TEMPERATURE RESIDUE SERVICE

- 3.1 Gland Packing Material: High-purity flexible graphite with Inconel wire reinforcement.
- 3.2 Live-Loaded Belleville Springs: Check spring deflection every 6 months to maintain load.